Prime Patterns in Five Computed Pictures: the local law, the spectrum of zeros, and exactly what remains open

Lee Smart*
Institute of Vibrational Field Dynamics
contact@vibrationalfielddynamics.org
@vfd_org

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Abstract

The ledger paper in this repository reports that ten prime-distribution phenomena pass pre-set numerical gates; it does not explain *why*. This companion supplies the why, at the level where the why is actually known, in five computed pictures. (1) Patterns of primes live or die cell-by-cell on each prime's *wheel*, and the surviving fractions multiply into the singular-series constants of Hardy and Littlewood's *Partitio Numerorum III*. (2) Those constants, times the standard density integral, predict the twin count below 5×10^7 to 0.003%, the relative abundance of all 105 even gaps to 210 within 0.31% (eight ledger-gated gaps within 0.25%), and Goldbach counts per individual N to 0.65% mean. (3) In the distributional sense of the explicit formula, the prime powers and the zeta zeros are a Fourier pair: we plot the primes' tapered spectrum, and every one of the 13 zeros in the plotted range aligns with a local peak — the zeros having been computed independently from ζ itself. The Chebyshev race makes the same interference audible in a single statistic. (4) Under the Riemann Hypothesis the zeros form a one-dimensional quasicrystal in Dyson's generalized sense — a pure point measure whose Fourier transform has pure point singular part, with atoms exactly at the prime-power logarithms plus an explicit smooth background — and Dyson proposed classifying such quasicrystals as a route to RH; the golden-ratio Fibonacci chain pictured here is the best-charted territory of exactly that catalogue. (5) Each open question met along the way — twin and Polignac infinitude, binary Goldbach, Cramér's bound, the unconditional Chebyshev density — is a statement about the zero distribution of an explicit L-function: not one object, but one *kind* of object, reached through one identity. All mathematics here is classical and attributed; the computations, figures, and their code are ours and ship with the repository. Nothing below proves, or claims to prove, any open conjecture.

*Companion to *The Prime Field Through One Instrument* (the verification ledger) in this repository: `github.com/vfd-org/closure-positivity-lab`. That paper verifies; this one explains. Every figure here is computed, not drawn — the generating script is `figures/make_pictures.py`, and two of the pictures are interactive at `explorables/`.

1 The wheels: where the densities come from

Look at the whole number line through a wheel with three cells — the residues $0, 1, 2$ modulo 3. Every integer lands on one cell; primes (other than 3 itself) only ever land on cells 1 and 2, because cell 0 means "divisible by 3."

Now place a *pattern* on the wheel instead of a single number: the twin pattern $(n,\, n{+}2)$. If n is on cell 0, the pattern dies (n is divisible by 3). If n is on cell 1, then $n+2$ is on cell 0: dead again. Only n on cell 2 survives. **One cell of three.**

Compare with what independence would give: two unrelated numbers each avoid cell 0 with probability $2/3$, so independent survival would be $(2/3)^2 = 4/9$. The twin pattern's actual survival is $1/3 = 3/9$. The wheel taxes twins by a factor

$$\frac{1/3}{(2/3)^2} \;=\; \frac{3}{4} \;=\; \left.\frac{p(p-2)}{(p-1)^2}\right|_{p=3},$$

and the same computation on every prime wheel $p > 2$ gives the factor $p(p-2)/(p-1)^2$: the pattern occupies two cells, so it dies on two of p, surviving $p-2$, against an independence baseline of $((p-1)/p)^2$. Multiply over all odd wheels:

$$\prod_{p>2} \frac{p(p-2)}{(p-1)^2} \;=\; \prod_{p>2}\Big(1 - \frac{1}{(p-1)^2}\Big) \;=\; C_2 \;=\; 0.6601618\ldots$$

This is the twin-prime constant of Hardy and Littlewood [1], and now you know what it *is*: the total tax the wheels charge the twin pattern, relative to independence. The predicted count below x is $2C_2 \int_2^x dt/\log^2 t$, and the sieve agrees to 0.003% at $x = 5 \times 10^7$ (ledger row 1).

Two more readings of the same picture. **Why gap 6 doubles:** the pattern $(n,\, n{+}6)$ doesn't move on the wheel of 3 (adding 6 is adding 0), so it dies only on cell 0: two cells of three survive — *twice* the twin survival, and no other wheel distinguishes the two patterns. So the predicted constant is exactly twice the twin constant, with no room for adjustment anywhere; the sieve ratio below 5×10^7 is 1.9950. **Why $(n, n{+}2, n{+}4)$ is impossible:** on the wheel of 3 the three offsets cover all three cells, so every starting position dies. The only escape is the pattern that *includes* 3 itself: $(3, 5, 7)$, once, never again. The ledger checks the analogous exact fact on the wheel of 5: the only twin pair $(p, p{+}2)$ with $p \equiv 3 \pmod 5$ is $(3, 5)$, because cell 3 forces $5 \mid p + 2$.

Said precisely: for a pattern of k offsets, the product of survival ratios over all wheels is its *singular series* $\mathfrak{S}$, and the Hardy–Littlewood k-tuple conjecture asserts that the count of occurrences below x is asymptotic to $\mathfrak{S} \int_2^x dt/\log^k t$ — the independence-model density corrected by exactly the wheel tax. The wheels themselves prove two things unconditionally: *admissibility* (no screen forbids the pattern) and the *value of the constant*. The asymptotic itself is conjectural for every $k \geq 2$; what the ledger verifies is that the counts track the conjectured formula at the 10^{-3}–10^{-2} level across every pattern tested. In the programme's language each wheel is a closure condition at the place p — a finite cyclic screen — and $\mathfrak{S}$ is an all-finite-places product: the decidable content lives place-by-place, the open content in how infinitely many places conspire.

2 One multiplication, every pattern

The wheel computation generalises mechanically: for a pattern of k offsets, count surviving cells $w(p)$ on each wheel, divide by the independence baseline, multiply. The result (the *singular series*) predicts, with *no further input*:

- the relative abundance of every even gap (figure below: all 105 gaps to 210, predictions as dots, sieve counts as bars — worst disagreement 0.31%, at gap 112; the eight ledger-gated gaps agree within 0.25%);
- Goldbach counts per individual N — the wheels of the primes *dividing* N relax their conditions, so N divisible by 3 has about twice the representations of its neighbours (verified: 200 consecutive even N near 10^6, mean deviation 0.65%);
- the twin count itself, to 0.003%.

What the wheels do *not* give is infinitude. The Hardy–Littlewood conjecture asserts that every admissible pattern's count keeps tracking $\mathfrak{S} \int dt/\log^k t$ forever; the wheels prove the admissibility and pin the constant, but the asymptotic is unproven for every single pattern — including twins, where the strongest unconditional statement is that *some* gap ≤ 246 recurs infinitely often [7, 8]. That question changes character entirely, and needs the second half of the story.

3 The duality: prime powers and zeros as a Fourier pair

Riemann's explicit formula is an exact identity. In von Mangoldt's form,

$$\psi(x) \;=\; x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log 2\pi - \tfrac{1}{2}\log\bigl(1 - x^{-2}\bigr), \qquad \psi(x) = \sum_{p^m \leq x} \log p,$$

valid for non-integer $x > 1$ with the zero sum taken symmetrically: the count of prime powers *is* a main term plus one oscillation per non-trivial zero ρ. Because the identity is exact it can be read in either direction, and the reverse reading is the physical one. Place the prime powers on a logarithmic line and let each broadcast a wave of frequency $m \log p$; sum the waves with $\sqrt{}$-damping and a smooth taper, and take the spectrum — the operation a crystallographer performs when she reads a diffraction pattern. The explicit formula says the spectrum must concentrate at the ordinates of the zeros.

The picture shows what the formula states: in the distributional sense of the explicit formula — prime powers included, suitable test functions understood — **the prime powers and the zeta zeros are a Fourier pair**. The identity runs both ways: zeros reconstruct prime counts, primes reconstruct zeros (the companion paper's prime-wave reconstruction of the critical line is this same duality run in reverse for our icosian L-function). Every question the wheels cannot answer is a question about this spectrum: the error in the prime count is a sum over zeros; the sharpest analytic handle on twin-prime infinitude is the correlations *between* zeros [9]. The wheels are the local data, the spectrum is the global data, and the explicit formula is the exact dictionary between them.

One global effect is visible without any spectral plot. Among the Λ-weighted prime powers mod 4, the *squares* of odd primes all land in the residue class 1; the explicit formula transfers that structural surplus into a deficit of actual primes $\equiv 1 \pmod 4$ of size $\sim \sqrt{x}/\log x$ — Chebyshev's bias, an interference fringe in a single statistic. The sieve shows the class $3 \pmod 4$ leading at 99.93% of all primes below 5×10^7, with the first lead change at exactly $x = 26{,}861$ [10]:

4 The crystal question: where it becomes geometry

A crystal produces sharp diffraction peaks because it is periodic. In 1982 Shechtman saw sharp peaks with a symmetry no period allows; the resolution is the *quasicrystal*, and its one-dimensional archetype is the Fibonacci chain — two tile lengths in ratio φ (the golden ratio), arranged by the substitution $A \to AB$, $B \to A$, never repeating.

Now make the connection precise, because it can be. A quasicrystal, in the sense Dyson uses, is a pure point measure whose Fourier transform is also (essentially) pure point. Write each non-trivial zero as $\rho = \frac{1}{2} + i\gamma_\rho$ and consider the measure $\sum_\rho \delta_{\gamma_\rho}$ on the spectral parameters $\gamma_\rho = (\rho - \frac{1}{2})/i$. The Weil form of the explicit formula computes its Fourier transform exactly: an explicit smooth background (from the Γ-factor and the pole at $s = 1$) plus *atoms at* $\pm m \log p$ — so the singular part of the transform is supported precisely on the logarithms of the prime powers. *If* the Riemann Hypothesis holds, every γ_ρ is real, the measure lives on the real line, and so: **under RH, the zeta zeros form a one-dimensional quasicrystal** in this generalized (Fourier-quasicrystal) sense — Dyson's own formulation [2], made distribution-theoretically careful by later authors. That conditional statement is precise mathematics, not analogy. Dyson's proposal is the converse strategy: develop a classification of one-dimensional quasicrystals, and identify the zeros within the catalogue — which would prove RH.

What is the catalogue's known territory? Precisely structures like the top panel: cut-and-project sets, of which the golden-ratio Fibonacci chain is the archetype, with Penrose tilings and the icosians of this repository as its higher-dimensional relatives [3]. That is the factual content of the pairing above — the best-charted region of the catalogue Dyson proposes to search is golden-ratio geometry — and it is the one place where the deepest open question about primes is, verbatim, a question about aperiodic order. The classification itself does not exist; nobody, including us, claims it; the honest status of every approach this repository has tested is recorded in the companion papers, negative results included.

5 The wall, and the map

Assemble the picture. Everything decidable about prime patterns we have met lives *locally*: wheels, survival counts, one multiplication — verified across the ledger's ten phenomena at levels from 0.003% (twins) to 0.65% (Goldbach mean), with the deterministic anchors (Leech's 26,861; the maximal gap 220 after 47,326,693) exact and three theorem-grade rows calibrating the instrument. Everything open lives *globally*, and while it is not literally one object, it is one *kind* of object: a statement about the zero distribution of an explicit L-function, reached through the explicit formula. Twin and Polignac infinitude have their sharpest analytic connection in the correlations of ζ's zeros [9, 4]; binary Goldbach's minor arcs and the unconditional Chebyshev density are controlled by zeros of Dirichlet L-functions [6]; the known progress toward Cramér's bound runs through zero-density estimates; the Sato–Tate rate through symmetric-power L-functions. Ledger row 10 measures the central case directly: spacing statistics of 1,983 computed ζ-zeros match the random-matrix prediction within the documented finite-height corrections [11, 5], with level repulsion suppressed $70\times$ relative to Poisson.

	the local layer (wheels)	the global layer (spectrum)
what it fixes	admissibility + the density constant	infinitude; error terms; extremes
status	decidable, computed, verified	open in every conjectural case
the "how"	counting cells on finite screens	open; framing: zero distributions
where geometry enters	each place is a finite cyclic screen	Dyson's quasicrystal proposal

That is the picture, as honestly as we can paint it: a local mechanism that is completely understood and verified to a fraction of a percent; a family of open global questions, each a statement about the zeros of an explicit L-function; an exact dictionary between the two layers; and one precise place where the open half becomes a question about aperiodic geometry. The ledger paper carries the gates, artifacts and falsifiers behind every number quoted here; the explorables let you drive both layers yourself.

References

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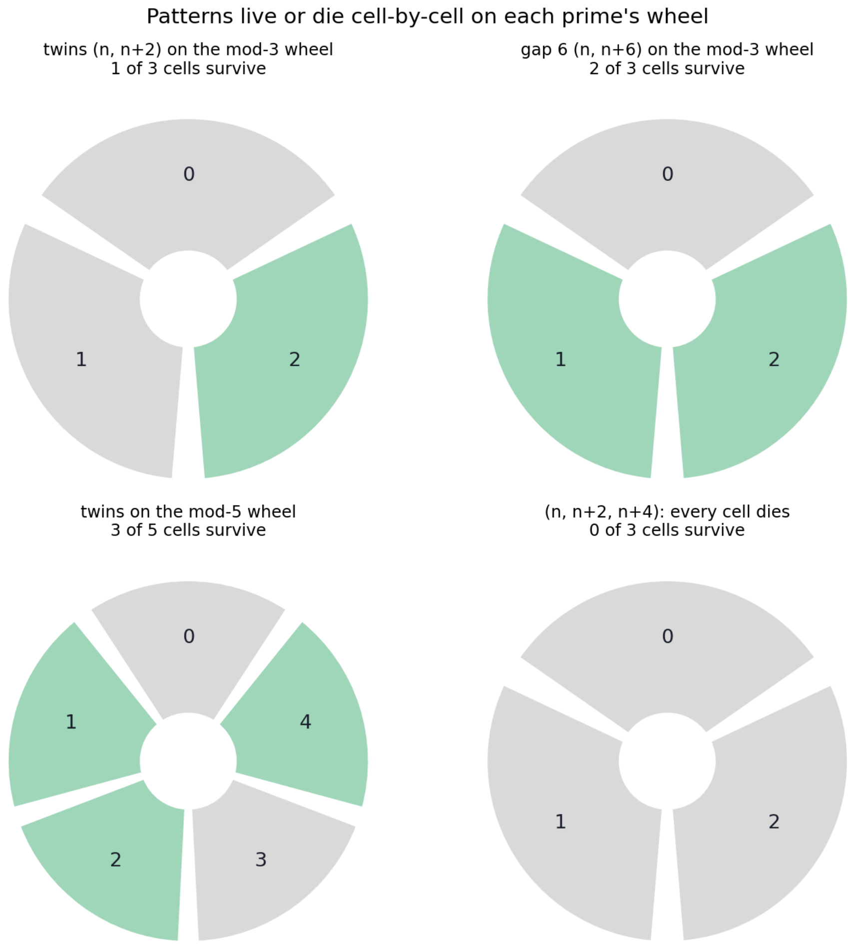

Figure 1: The mod-3 and mod-5 wheels. Green cells: starting positions where the pattern avoids every forbidden cell. Twins survive 1 cell of 3 (top left); the gap-6 pattern survives 2 of 3 (top right) — that single picture is the whole reason the Hardy–Littlewood prediction for gap-6 pairs is exactly twice the twin prediction (the sieve ratio: 1.9950). Bottom right: $(n, n+2, n+4)$ kills every cell mod 3, so only one such triple exists in all of mathematics — $(3, 5, 7)$, where the dead cell is the prime 3 itself.

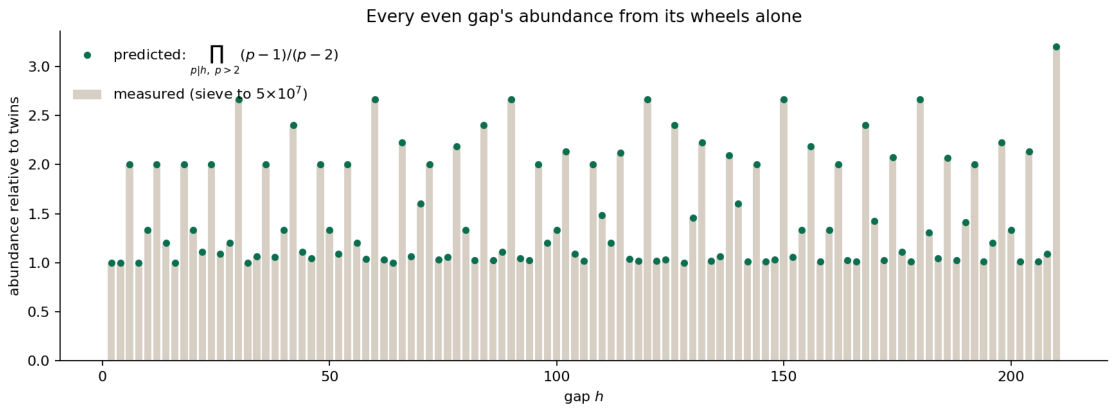

Figure 2: All even gaps to 210: measured abundance (bars) against the wheel product (dots). The spikes are gaps divisible by 3, 15, 105… — more surviving cells, more pairs. Interactive version: `explorables/gap-slider.html`.

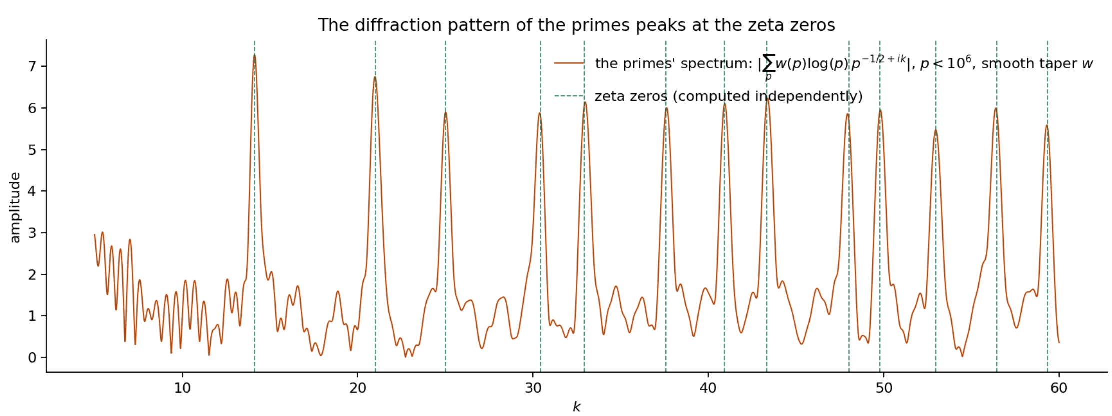

Figure 3: The spectrum of the prime waves (orange), computed from the primes below 10^6 with a smooth (Riesz) taper, and *nothing else*; dashed lines are the zeta zeros, computed independently in PARI/GP from ζ itself. Every one of the 13 zeros in the plotted range aligns with a local peak within 0.15. (A sharp cutoff instead of the taper Gibbs-rings and aligns only 5 of 13 — the taper choice is documented in the generating script.)

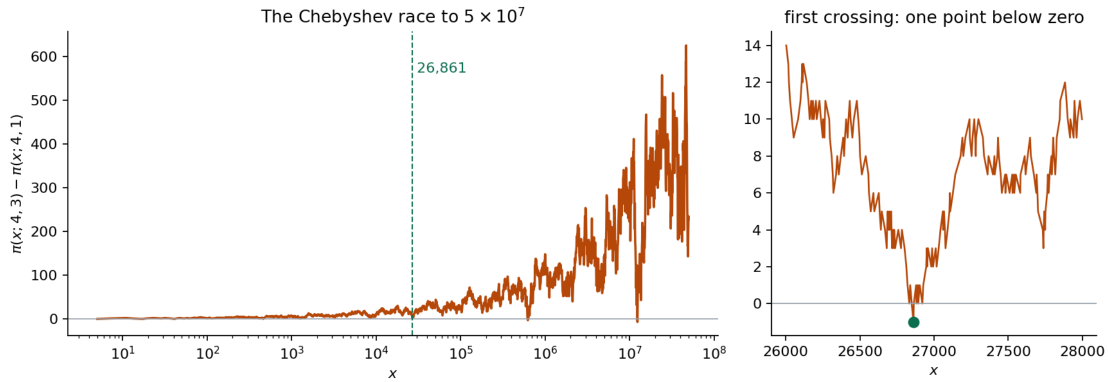

Figure 4: The Chebyshev race $\pi(x; 4, 3) - \pi(x; 4, 1)$, computed exactly. Left: the full race to 5×10^7 on a logarithmic axis. Right: the first crossing — a single point dips below zero at $x = 26{,}861$, then the lead recovers for roughly the next 47,000 primes, until the second reversal zone near $x = 616{,}841$. Under GRH plus linear independence of the zeros, the lead set has logarithmic density $0.9959\ldots$ [6]; unconditionally, even the existence of that density is open.

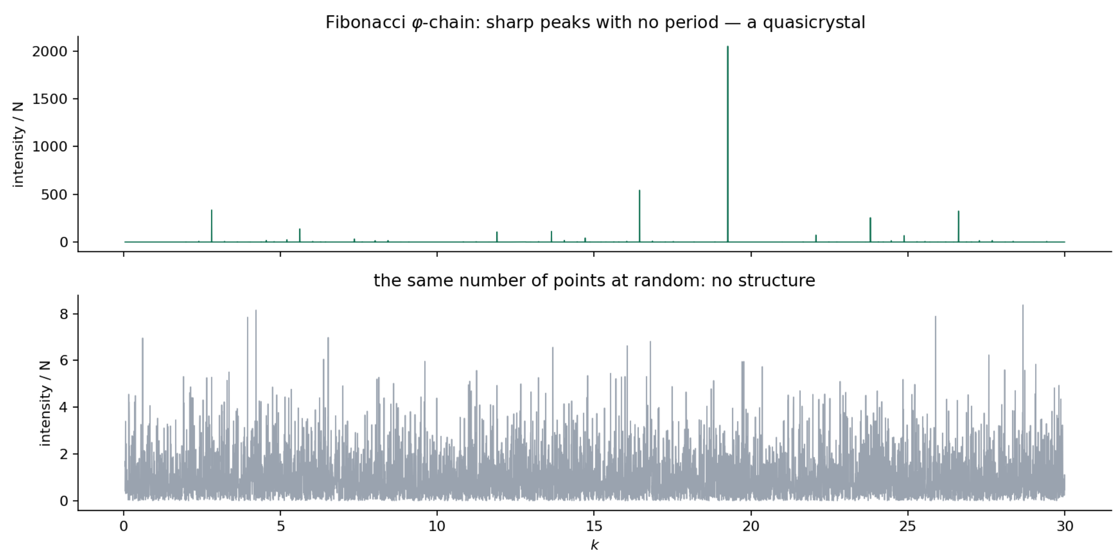

Figure 5: Computed diffraction of 3,000 points. Top: the Fibonacci φ-chain — razor-sharp peaks, no period. Bottom: the same number of points placed at random — featureless. Sharp spectrum without periodicity is possible, and golden-ratio geometry is its canonical home.